

Evidential Bayesian Neural Networks for Uncertainty-Aware Classification

John-Pierre Weideman
Master's Thesis Defence
TODO: defence date

Showcase
Figure

- 1 Complementary Representations of Predictive Uncertainty
- 2 Categorical Evidential Bayesian Neural Network
- 3 Dirichlet Multinomial Evidential Bayesian Neural Network
- 4 Implications and Conclusion

Message. A class label alone is not enough. We need the predictive uncertainty and its sources.

TODO: content. Three uncertainty terms, one line each. Entropy decomposition in one line.

Message. The Cat-BNN represents aleatoric and epistemic uncertainty. It represents no distributional uncertainty.

TODO: content. Weight posterior, prediction $f(x; w)$, point mass on π , zero distributional term.

Message. EDL represents aleatoric and distributional uncertainty. It represents no epistemic uncertainty.

TODO: content. Concentration parameters $\alpha(x; w)$, Dirichlet over π , one trained weight configuration.

Message. We combine the weight posterior with the Dirichlet network. The result represents all three uncertainty terms.

TODO: content. Table of models against uncertainty terms. Cat-eBNN for single labels. DM-eBNN for repeated labels.

- 1 Complementary Representations of Predictive Uncertainty
- 2 Categorical Evidential Bayesian Neural Network
- 3 Dirichlet Multinomial Evidential Bayesian Neural Network
- 4 Implications and Conclusion

Message. The Cat-eBNN is the Dirichlet network of EDL with a posterior over the shared network weights.

TODO: content. Model sketch. Likelihood is the Dirichlet mean $m_y(x; w)$.

Message. At a fixed mean vector, a larger total concentration parameter gives a higher aleatoric term and a lower distributional term.

TODO: content. Figure of both terms against α_0 at a fixed mean vector. New figure from scripts/.

Message. The single label likelihood is invariant to the total concentration parameter at a fixed mean vector. So we state the assumption with a Gamma prior.

TODO: content. Scale invariance in one line. Gamma prior with prior mode and prior standard deviation. Honest scope sentence.

Message. We compare the Cat-eBNN and the Cat-BNN on Fashion-MNIST and CIFAR-10 with SGLD.

TODO: content. Datasets. One SGLD chain, fixed budget. Cold, warm, shifted initialisation. Prior sweeps. Metrics.

Message. Warm and shifted initialisation give higher validation accuracy than cold initialisation. Shifted initialisation avoids the accuracy loss when the prior mode is far above the checkpoint.

TODO: content. Trajectory figure or table extract. One line on finite sampling budget.

Message. The realised total concentration follows the prior mode. A narrower prior keeps it closer. A higher realised total concentration gives a higher aleatoric share and a lower distributional share.

TODO: content. Extracts of the mode sweep and sd sweep tables. One line on the fixed mean effect.

Message. The Cat-eBNN has competitive accuracy but no consistent predictive improvement over the Cat-BNN. Its contribution is the complete uncertainty representation and the explicit prior assumption.

TODO: content. Extract of the predictive table. Two closing lines on contribution and scope.

- 1 Complementary Representations of Predictive Uncertainty
- 2 Categorical Evidential Bayesian Neural Network
- 3 Dirichlet Multinomial Evidential Bayesian Neural Network**
- 4 Implications and Conclusion

Message. Repeated labels per input, aggregated by class into count vectors, carry information that single labels do not.

TODO: content. One CIFAR-10H image with its count vector n and count total R .
Definition in one line.

Message. The DM-eBNN models count vectors with a Dirichlet multinomial likelihood. The M-BNN is the Bayesian baseline with a multinomial likelihood.

TODO: content. The two observation models side by side.

Message. At a fixed mean vector and a count total above one, the count variance depends on the total concentration parameter. So the count likelihood can inform it. The M-BNN has a fixed count variance.

TODO: content. Dirichlet multinomial variance in one line. Variance ratio. The Gamma prior stays part of the posterior.

Message. We compare the DM-eBNN and the M-BNN on CIFAR-10H.

TODO: content. Two initialisations, two prior modes. Mean matched controls. Count NLL and majority label accuracy.

Message. The DM-eBNN has lower count NLL than the M-BNN under both initialisations and both prior modes, with competitive majority label accuracy. The mean matched controls show that the additional count variance drives the better fit.

TODO: content. Extract of the count prediction table.

Message. The realised total concentration follows the prior mode and the initial checkpoint within the sampling budget. Every cold initialisation fit without the Gamma prior reaches the boundary $\alpha_0 = C$.

TODO: content. Extract of the prior sensitivity table. Optionally one panel of the cold collapse figure.

Message. The DM-eBNN reports all three uncertainty terms, and count data inform the allocation between aleatoric and distributional uncertainty. The interpretation and practical value are future work.

TODO: content. Extract of the count terms table.

- 1 Complementary Representations of Predictive Uncertainty
- 2 Categorical Evidential Bayesian Neural Network
- 3 Dirichlet Multinomial Evidential Bayesian Neural Network
- 4 Implications and Conclusion**

Message. The eBNNs make the allocation assumption explicit. Single labels cannot inform it, count data can. This motivates collecting and retaining repeated labels.

TODO: content. Three lines. Cat-BNN implicit point mass. EDL allocation from the training objective. The eBNNs state it with a Gamma prior, and the DM-eBNN lets count data inform it.

Message. Two models represent all three uncertainty terms. The DM-eBNN also improves count prediction. Open points remain.

TODO: content. Summary in three lines. Limits and future work in three lines. This slide stays on screen during questions.

TODO: content. The Cat-eBNN against earlier Bayesian Dirichlet combination models. The DM-eBNN against models for repeated labels.

TODO: content. Aleatoric term in closed form. Distributional term as the remainder.

TODO: content. Rescaling $\alpha(x; w)$ leaves $m(x; w)$ unchanged. The single label likelihood is a product of mean components.

Backup. The Gamma Prior

TODO: content. Density, prior mode, prior standard deviation.

TODO: content. Marginal likelihood loss is invariant to $\alpha_0(x; w)$. The KL term sets it.

Backup. SGLD and the Sampling Budget

TODO: content. Sampler, one chain per run, budget, checkpoint restore.

TODO: content. One frame per full table as needed.

TODO: content. Per seed tables for single labels and repeated labels.

Backup. Cold Initialisation without the Gamma Prior

TODO: content. The four panels of the cold collapse figure.